Indian Academy of Sciences, Bengaluru
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SUMMER RESEARCH FELLOWSHIPS — 2021

Format for the Four-week Report*,^

Name of the candidate	:	ADHIRAJ BANERJEE
Application Registration no.	:	ENGS589
Date of Commencement of work	:	22/06/2021
Mode of work	:	From Home: ✓ Guide's Laboratory:
Name of the guide	:	Dr. SIVA RAM KRISHNA VADALI, Senior Principal Scientist
Guide's institution	:	Central Mechanical Engineering Research Institute, Durgapur
Place of stay during the tenure of the fellowship (if working in guide's institution)	:	Hostel provided by — Guide — Own arrangement — Other (Specify) Internship carried online

Adhiraj Banerjee
Signature of the candidate
Date: 25/07/2021

Signature of the guide 26/07/2021
Date:

	INSPIRE/KVPY FELLOWSHIP (please fill this box)#		
1.	I am currently a recipient of	INSPIRE FELLOWSHIP	Yes / ✓ No
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		If, YES, fill cols. 2, 3 & 4	
2.	**INSPIRE/KVPY** Fellowship is from _____[month]/_____[yr] to _____[month]/_____[yr]		
3.	I receive a monthly fellowship of Rs. _____ from INSPIRE/KVPY towards my living expenses		
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	I affirm that the information given above is correct.	Adhiraj Banerjee **Signature of the candidate**	

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Candidate's name:		Fellowship amount:
Student:	Teacher:	Deduction:
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IMPORTANT NOTES:

* The four-week report could be between 300 and 350 words.

^ This format should be the first page of the report and should be stapled with the main report.

\# Mandatory to fill this section, this should be filled and signed by you even if you are not an INSPIRE/KVPY Fellow. Otherwise release of fellowship amount will be withheld.

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Efficacy of Gaussian Mixture Model in a Machine Learning Framework for Speaker Recognition
A Four Week Report by

Adhiraj Banerjee (SRFP Application No. : ENGS589)

July 25, 2021

1 Introduction

Speaker Recognition (SR) has found numerous applications such as voice biometric, forensics, traitor finding, voice-mail and tele-banking. Conceptually, SR determines the presence of an active speaker among a set of registered speakers. In view of the applicability of Machine Learning framework to a wide variety of classification problems, in the present work emhasis is laid on the popular Gaussian Mixture Model (GMM) based ML (GMM-ML) approach as a potential solution for SR. Voice samples of ten different speakers are recorded in a python enviroment and the performance of GMM-ML for SR is analyzed. It is observed that GMM-ML based SR is highly accurate in distinguishing the actual speaker from the 10 registered speakers.

2 "Machine Learning" for Speaker Recognition

Machine Learning (ML) is a mathematical framework which helps systems to learn from trained model and predict the class in which the test data may be classified into. The present work studies the efficacy of a ML algorithm for Speaker Recognition. Going by principles of ML, a model for Speaker Recognition is trained such that the model identifies unique patterns (referred as features) from a set of different voice samples (of registered speakers) and distinguish the actual speaker voice sample from the rest. In the present work, a familiar ML algorithm, Gaussian Mixture Model (GMM) based classification, is chosen and its efficacy in accurate speaker recognition is studied.

2.1 Gaussian Mixture Model based SR

It is known that the Gaussian Mixture Model (GMM) is a powerful model used to solve clustering problems. It features a probabilistic approach by which it

approximates the N dimensional input data as a linear combination of N Gaussian distributed random variables with mean vector $\boldsymbol{\mu}$=$\{\mu_i\}_{i=1,2,\ldots,N}$ (i.e. order $NX1$), variance-covariance matrix $\boldsymbol{\Sigma}$ (i.e. order NXN) and $\boldsymbol{\omega}$=$\{\omega_i\}_{i=1,2,\ldots,N}$ (i.e. order $NX1$) as the weights in the linear combination. In *Speaker Recognition*, features of speaker's voice sample are extracted into a feature vector and used as data points for processing to identify the registered speaker. Since the feature vector is *multi-dimensional*, the probability density function (PDF) of particular class of speaker (λ) can be expressed as a linear combination of N multivariate Gaussian PDFs as:

$$P(\boldsymbol{X}|\lambda) = \sum_{i=1}^{N} \omega_i \cdot p_i(\boldsymbol{X}|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i), \tag{1}$$

where $-$ K denotes the number of Gaussian random variables used for N clusters formed from the input vector $\boldsymbol{X}$; λ, as mentioned earlier, is the class to which a speaker is associated with; $\boldsymbol{X}$ represents the N dimensional training data; and $p_i(\boldsymbol{X}|\mu_i, \boldsymbol{\Sigma}_i)$ is the multivariate Gaussian PDF given by,

$$p_i(\boldsymbol{X}|\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i) = \left((2\pi)^N \cdot |\boldsymbol{\Sigma}_i|\right)^{-0.5} \cdot exp\left(-0.5(\boldsymbol{X} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1} (\boldsymbol{X} - \boldsymbol{\mu}_i)\right). \tag{2}$$

In the formulation in Equation 1, the probability, also referred as the likelihood, computed from the input data points in vector X is assigned to a particular class corresponding to the speaker's GMM. Herein, the mean vectors ($\boldsymbol{\mu}_i$), variance-covariance matrices ($\boldsymbol{\Sigma}_i$) and weights of each of the gaussian components ($\boldsymbol{\omega}_i$) is updated iteratively till the likelihood for the data points (i.e. feature vectors) is maximized, thereby creaing a trained model with the extracted features. The maximized likelihood will thus be assigned to the class corresponding to the speaker, in turn creating the the classified GMM for the speaker.

2.2 Feature Extraction in SR: Mel-Frequency Cepstrum Coefficients(MFCC)

Mel-Frequency Cepstrum Coefficients (MFCCs) are prominent features used for feature extraction and data train in speaker recognition. A principle reason for using MFCCs is due to the fact that they provide an clear idea about the perceived difference between frequencies, especially in high frequency region. Specifically, the importance of MFCCs arises from the fact that *the human ear can perceive voice signal frequencies in a non-linear fashion.* From here, it is evident that the audio frequencies needs to be analyzed in the *Mel* scale, rather than the Hertz scale. There is logarithmic (i.e. non-linear) relationship between the Mel scale and Hertz scale, from which frequencies in Mel scale becomes more useful. In the following, the extraction of MFCCs is discussed in brief:

- First, Pre-Emphasis is performed to increase the energy of the signal at higher frequencies, i.e. the frequency range where the human ear tends to

perceive less in linear scale. Also, windowing is performed to remove discontinuities or overlapping leading to loss of information between frames of the audio signals.

- Next, N-point Discrete Fourier Transform (DFT) of each frame of the signal is obtained.
- The frequency spectrum is then passed through a Mel-filter bank, which is a bank of triangular filters plotted against frequencies in Mel Scale. The filter output we get after passing the frequency spectrum into Mel-filter bank gives the Mel-Frequency Spectrum. The spectrum obtained from filtering operation renders non-linear frequency spectrum at the higher frequencies.
- Next, the logarithm of the Mel-frequency spectrum is taken to obtain Mel-Frequency Cepstrum.
- Finally, Cepstrum coefficients are operated upon by taking its Discrete Cosine Transform which provides the desired MFCCs.

3 Results and Inference

In the present work, GMM based Speaker Recognition is tested for its efficacy over 10 speakers. Note that the programming is performed in a Python environment. It is observed that GMM based SR is highly accurate (99% approx).

4 Conclusion on Work Done and Further Work

Speech Recognition is a *text-dependent* method, where it highly depends on the language and corpus. On the other hand, Speaker Recognition mainly focusses on raw audio percepts (and information derived therein) to identify the uniqueness aspect among different speakers. *Machine Learning* is a holistic approach which renders *Speaker* Recognition efficient when compared to *Speech* Recognition methods for Speaker idenfitication applications. In this work, GMM based ML is applied for classification in a speaker recognition problem. It is found that an accuracy as high as 99% is achieved when tested over 10 different speakers. Further work in this interesting research area is to analyze the accuracy of GMM based ML when the voice recorded by the speaker is either noisy or when multiple speakers are active.